

Yu Li

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Education

George Washington University **Washington, D.C.**
Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0 Aug 2025 – May 2029
 Advisor: [Prof. Tian Lan](#) 🌐; Mentor: [Prof. Zhengling Qi](#) 🌐 (expected)

Wuhan University, Hongyi Honor College **Wuhan, China**
B.Eng. in Microelectronics Science and Technology, GPA: 3.87/4.0 Sept 2021 – Jun 2025
 Advisor: [Prof. Cheng Lei](#) 🌐 & [Prof. Wei Liu](#) 🌐

Research Interests

My research focuses on **LLM post-training**, **agent policy learning**, and **generative AI**. Specific topics include LoRA fine-tuning, preference optimization, reasoning, agentic RL, and multimodal in-context learning.

Industry Experience

ByteDance Inc. **San Jose, CA**
Machine Learning Scientist Intern, Recommendation Core Data May 2026 – Aug 2026
 Worked on ML-driven video recommendation for TikTok's production system, focusing on content understanding and LLM-based ranking within a cross-functional team.

Preprints & Under Review

- [P.1] [OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning](#) 🌐
 Yu Li, Rui Miao, Tian Lan, Zhengling Qi
- [P.2] [ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning](#) 🌐
 Yu Li, Rui Miao, Zhengling Qi, Tian Lan
- [P.3] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**
 Yu Li, Binxu Li, Tian Lan
Under review at COLM, Avg rating: 6.5
- [P.4] [Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO](#) 🌐
 Yu Li, Tian Lan, Zhengling Qi
- [P.5] [INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization](#) 🌐
 Yu Li, Tian Lan, Zhengling Qi
 AI with Recursive Self-Improvement Workshop at ICLR, 2026

Publications

C=Conference, J=Journal, †=Equal Contribution

- [C.1] **CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion**
 Yu Li, Yujun Cai, Chi Zhang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

[C.2] Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization

Yu Li, Sizhe Tang, Tian Lan

Findings of the Association for Computational Linguistics (ACL), 2026

[C.3] KG-SAM: Injecting Anatomical Knowledge into Segment Anything Models via Conditional Random Fields

Yu Li, Da Chang, Xi Xiao

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) [Oral], 2026

[C.4] ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense

Yu Li†, Sizhe Tang†, Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan

Intelligent Cloud Computing and Networking Workshop at INFOCOM, 2026

[C.5] Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT

Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang

The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026

[C.6] Mixed Text Mode Optical Character Recognition Based On Transformer

Yu Li†, Da Chang†

International Conference on Neural Information Processing (ICONIP), 2024

[J.1] Dual branch SAM-Transformer Fusion Network for Accurate Breast Ultrasound Image Segmentation

Yu Li, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

Medical Physics, 2025

[J.2] SfMDiffusion: Self-supervised Monocular Depth Estimation in Endoscopy Based on Diffusion Models

Yu Li, Da Chang, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

International Journal of Computer Assisted Radiology and Surgery, 2025

[J.3] Local Optimum Time-Reassigned Synchrosqueezing Transform for Bearing Fault Diagnosis

Site Lv, Shan Zeng, Yu Li, Ke Yang, Yulong Chen

IEEE Sensors Journal, 2024

Selected Awards & Honors

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- **ICML 2026 Gold Reviewer**, top reviewers recognition. 2026
 - **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. 2024
 - **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. 2023, 2024
 - **First-Class Scholarship**, Top 5%, ¥3,000, three times. 2022, 2023, 2024

Academic Services

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- **Conference Reviewer:** {NeurIPS, COLM, ACM MM, ICML, ICLR, AAAI, ICASSP}'2026
 - **Journal Reviewer:** TMLR, TPAMI, ToN, Neurocomputing

Skills

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- **Languages:** English (TOEFL 105), Chinese (Native), Japanese (Basic)
 - **ML/LLM Stack:** PyTorch, DeepSpeed, Flash Attention, vLLM, OpenRLHF, VERL, TRL
 - **Tools & Platforms:** Linux/Ubuntu, Docker, Git, Weights & Biases, Cadence